Australia: energy security, economic opportunity and partnership priorities

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AI-generated from the data and checked against it; paper baseline as of 12 October 2026.

Summary

- Australia is most exposed in midstream wind, midstream solar and midstream electric grid, where its Energy Security is higher than only 22%, 26% and 40% of the other economies scored. Trade deficits and small export shares pull each down.[1]
- Its strongest openings are in upstream electric vehicles, upstream batteries and downstream batteries, where its Economic Opportunity is higher than 98%, 97% and 95% of the other economies scored. Investment pushes each of them up.[1]
- On the paper's weighting, the Strategic Index puts downstream batteries 1st among Australia's own sectors, a narrow lead that rests on policy activity; without the policy term, midstream electric grid ranks 1st.[1]
- The Partnership Strength Index rates South Korea highest for Australia, as an export market; it is also Australia's 4th trade partner. China ranks 1st among its trade partners but 36th on Partnership Strength.[1,2]

1. Energy security risks and why

Energy Security measures how well a country can rely on what it needs in a sector, compared with other economies. Of Australia's 28 placed sectors (those comparable on both scores), 4 sit in the two low-energy-security corners: all 4 in the Secure corner and none in the Upgrade corner. The Secure corner pairs low opportunity with low energy security. Compared with other economies in each sector, its Energy Security is lowest in midstream wind (0.38), midstream solar (0.33) and midstream electric grid (0.47), all in the Secure corner. It scores higher than only 22%, 26% and 40% of the other 64, 77 and 92 economies scored in these sectors.[1]

The trade category, which counts a country's trade deficit in these goods, its share of world exports and how concentrated world exports are, pulls each of these scores below the middle economy. With the Upgrade corner empty, its high-priority sectors are its leading Strategic Index sectors: downstream batteries, midstream batteries and midstream solar. In midstream solar, the deficit was 0.4% of GDP in 2023; on that measure Australia scores higher than only 27% of the other economies. For context, its largest suppliers of midstream solar goods in 2024 were China (53.2% of imports totalling USD 7.8 bn) and the United States (6.5%).[1,2,3]

Among fuels, upstream oil sits in the Watch corner; its Energy Security there is higher than 57% of the other 96 economies scored. Energy imports pull that score down: oil production minus consumption was -1.54 EJ in 2025. Proved reserves of 2.39 bn bbl (2020) push it up.[1,4]

2. Economic opportunities and why

Economic Opportunity measures how well placed a country is to create value in a sector, from exports and related capabilities to investment, costs and demand. Of the 28 placed sectors, 9 sit in the two high-opportunity corners, all 9 in the Grow corner, which pairs high opportunity with high energy security. Its Economic Opportunity ranks highest in upstream electric vehicles, upstream batteries, downstream batteries and downstream solar, higher than 98%, 97%, 95% and 94% of the other economies scored.[1]

Investment pushes the upstream scores up: on capacity operating today, Australia scores higher than 95% of the other economies in upstream electric vehicles and 93% in upstream batteries. Production also pushes both up: on growth in production it scores higher than 98% and 95% of the other economies. In downstream batteries, investment and consumption push the score up: it scores higher than 96% of the other economies on annual investment and 99% on use per person today. In downstream solar, production and energy access push it up: generation was 58.4 TWh and use came to 7.79 GJ per person, both in 2025.[1,4]

For context, China took 88.1% of its exports in upstream electric vehicles (USD 4.31 bn in 2024) and 90.6% of those in upstream batteries (USD 4.19 bn). The paper's framework associates the Grow corner with export promotion.[2]

3. Sectoral priorities and the existing policy posture

The Strategic Index ranks Australia's own sectors by combining opportunity, security need, policy activity, the supply-chain stage and the technology's low-emission value. On the paper's weighting, it puts downstream batteries 1st, midstream batteries 2nd, midstream solar 3rd, upstream batteries 4th and midstream electric grid 5th among its own sectors. At 0.62, downstream batteries leads midstream batteries by 0.005, and midstream batteries leads midstream solar by 0.006: both pairs are almost tied. Policy activity accounts for the lead of downstream batteries, adding 0.200 to its score. On data-based priority, the Strategic Index without its policy term, it ranks only 12th among Australia's own sectors; midstream electric grid, in the "middle" policy group, ranks 1st. In midstream solar, a Secure-corner sector, security need adds 0.200. Its allied pick, the top upstream or midstream sector after a resource screen, is midstream batteries.[1]

Policy activity counts International Energy Agency Policies and Measures records and reads the Climate Action Tracker rating against each technology. Splitting its sectors into thirds on each measure, top priority meets the most policy in midstream solar, midstream batteries, midstream wind, upstream batteries and upstream solar. No top-priority sector has the least policy and no bottom-priority sector the most; bottom priority meets the least policy in 6 sectors, every oil and coal sector among them. Australia carries the Climate Action Tracker rating Insufficient (June 2025) and has 584 Policies and Measures records, 392 of which name a technology. For context, Global Trade Alert lists 507 active industrial-policy measures for Australia, most often tied to midstream solar (122) and midstream wind (99).[1,5,6,7]

4. Partnership priorities and why

The Partnership Strength Index rates partners from Australia's viewpoint as reliable suppliers (Friendshore), as export markets (Partnership Opportunity) and, for middle- and lower-income partners, by Development Potential. It ranks South Korea 1st, Germany 2nd, France 3rd, Finland 4th and Japan 5th of 191 partners. South Korea stands out as an export market, with best sectors including upstream batteries, upstream electric vehicles, upstream gas, upstream wind and midstream nuclear. Germany, France and Finland each

stand out as a reliable supplier; for Germany the best sectors include midstream green hydrogen, midstream electric vehicles, midstream solar, midstream wind and midstream gas.[1]

Today's ties only partly follow these priorities. South Korea is also Australia's 4th trade partner on average across its energy-goods baskets, Japan its 6th and Germany its 7th. France, by contrast, is a priority partner with few ties today: it ranks 27th by trade. The United States, ranked 6th, is its 2nd trade partner and shares the Indo-Pacific Economic Framework, the Minerals Security Partnership and a U.S. free-trade agreement with it. Australia's firms have 4 cross-border factory projects there. China ranks 1st among its trade partners but 36th on Partnership Strength, and the countries vote in opposite blocs at the UN.[1,2,8,9,10]

As a partner to the coalition, Australia ranks 22nd of 192 economies on Partnership Strength averaged over the core members' viewpoints. In the allied network design, which spreads upstream and midstream production across coalition members, its largest designed shares are 22.0% of upstream batteries, 21.5% of upstream electric vehicles and 19.5% of upstream wind.[1,11]

5. Data notes and caveats

The Country Insights page shows these data as of 12 October 2026.

- Scores compare Australia with other economies in the same sector; they are not measures of risk or profit in absolute terms.
- All index values are the paper's baseline as of 12 October 2026; Strategic Insights recomputes them under other weightings.
- Strategic Index scores within 0.01 of each other are almost tied: small changes in the data or the weights can swap their order.
- Factory projects are cross-border solar, wind, battery and electric-vehicle manufacturing projects only, announced projects included, which may never be built; domestic investment is not covered (Cleantech Manufacturing FDI Dataset, edition 2026.10.05, CC BY 4.0; counts by the authors, a joint venture counted under its listed parent).
- Industrial-policy measures from the Global Trade Alert, New Industrial Policy Observatory, October 2026 extract (as of 5 October 2026) are context only, outside the Policy index.

Sources

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Allied Industrial Base Explorer, Country Insights: Australia